

A repeated interlace root below -4

Disproof of conjecture 00000003474

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Result. For the disjoint union $G = P_5 \sqcup P_5$ of two five-vertex paths, the original one-variable interlace polynomial is

$$q_N(G; x) = x^2(x^2 + 5x + 2)^2.$$

Consequently $r = (-5 - \sqrt{17})/2 < -4$ is a multiple real root. This contradicts the conjecture's first clause, that all multiple roots lie in $[-4, 0]$.

Definition and scope. We use the original single-variable interlace polynomial of Arratia, Bollobás, and Sorkin [1], subsequently called the *vertex-nullity interlace polynomial*. It is characterized on finite simple graphs by

$$q_N(E_n; x) = x^n, \quad q_N(H; x) = q_N(H - a; x) + q_N(H^{ab} - b; x) \quad (ab \in E(H)). \quad (1)$$

Here E_n is edgeless. For the ABS pivot H^{ab} , divide the other vertices into those adjacent to a only, b only, both, or neither; toggle edges between different classes among the first three. Incidences at a, b remain unchanged. The two source languages do not specify a different interlace variant or restrict to connected graphs. The counterexample uses this standard original convention, with no claim about a different polynomial.

Proposition 1. *The graph $G = P_5 \sqcup P_5$ has a multiple real interlace root outside $[-4, 0]$.*

Proof. Write $p_n = q_N(P_n; x)$, where P_n has n vertices and $p_0 = 1$. Pivoting on an end edge ab , with a the leaf, changes no edges. Deleting a leaves P_{n-1} ; deleting b leaves an isolated a and P_{n-2} . The polynomial is multiplicative on disjoint unions [1, Remark 16], giving $p_n = p_{n-1} + xp_{n-2}$ for $n \geq 2$. Starting with $p_1 = x$ gives

$$p_2 = 2x, \quad p_3 = x^2 + 2x, \quad p_4 = 3x^2 + 2x, \quad p_5 = x^3 + 5x^2 + 2x.$$

Multiplicativity now yields $q_N(G; x) = p_5^2 = x^2(x^2 + 5x + 2)^2$. The quadratic has root $r = (-5 - \sqrt{17})/2$. Since $17 > 9$, we have $\sqrt{17} > 3$ and hence $r < -4$. Its other root differs from r and $r \neq 0$, so r has multiplicity exactly two in $q_N(G; x)$. One false necessary clause already disproves the full conjecture; its additional embedding and link-spectrum clauses are not needed. $\square$

Lean verification. The accompanying Lean 4.19.0 project, with Mathlib pinned to commit c44e0c8ee63ca166450922a373c7409c5d26b00b, contains:

- actual Boolean adjacency matrices on $\text{Fin}(5)$ and $\text{Fin}(10)$, with symmetry and absence of loops checked;
- vertex deletion, the ABS pivot, and the total recursion (1), choosing the first edge in lexicographic order;
- kernel-checked evaluation of that recursion on both graphs, proving the two polynomial identities above;
- proofs that the ten-vertex polynomial is nonzero, that $(X - r)^2$ divides it over $\mathbb{R}$, and that $r < -4$;
- a theorem negating the universal real multiple-root location claim for this interlace polynomial.

To keep finite graph evaluation small, an auxiliary recursion records the orders of the edgeless leaves. A general Lean theorem proves that the sum of their monomials equals the polynomial recursion. Ordinary `decide` checks the finite reduction lists; polynomial algebra is proved by kernel-checked tactics. There is no use of `native_decide`, unfinished proof, or custom axiom.

Formalization boundary. The Lean definition directly implements the published original pivot recursion. The general theorem that its value is independent of pivot order, and its equivalence to the subset-nullity expression below, are the established results cited in [1, 2]; they are not reproved in this file. No hypothesis assigning a precomputed polynomial to G is assumed. The formal multiple-root certificate proves multiplicity at least two, which is all the disproof requires; the exact multiplicity statement follows from the elementary factorization in the paper.

Independent exact computation. The pure-Python check enumerates all induced vertex subsets and computes their adjacency-matrix ranks over $\mathbb{F}_2$, using

$$q_N(H; x) = \sum_{S \subseteq V(H)} (x - 1)^{\dim \ker A_H[S]}.$$

For P_5 , the numbers of subsets of nullities 0, 1, 2, 3 are respectively 8, 15, 8, 1. Expanding gives $x^3 + 5x^2 + 2x$. For G , all 1024 subsets are checked, giving ascending coefficients (0, 0, 4, 20, 29, 10, 1), exactly the square of the path polynomial. This computation is supplementary and is not imported as a Lean axiom.

References

- [1] R. Arratia, B. Bollobás, and G. B. Sorkin, *The interlace polynomial of a graph*, J. Combin. Theory Ser. B **92** (2004), 199–233. Definition 5, Theorem 12, and Remark 16. arXiv:math/0209045.
- [2] R. Arratia, B. Bollobás, and G. B. Sorkin, *A two-variable interlace polynomial*, Combinatorica **24** (2004), 567–584, Section 4. arXiv:math/0209054.